

AI Architectural Design Consultant: A Retrieval-Augmented Generation Framework for Context-Aware Conceptual Architectural Design

Abstract

While Generative AI is increasingly used for architectural concept design, standard text-to-image models often generate visuals that ignore structural reality, site context, and climate factors. Because these tools work primarily as basic image generators, their practical value for architects and students remains limited. In this project, we develop the AI Architectural Design Consultant, a system designed to create a more grounded and context-aware design workflow. Our approach combines Large Language Models (LLMs), FAISS vector databases, Retrieval-Augmented Generation (RAG), and diffusion models.

Instead of relying on an LLM's default training alone, our framework extracts spatial and environmental constraints from user prompts and actively retrieves real-world architectural knowledge and case studies. Using this retrieved evidence, the system generates structured design recommendations, sustainable climate strategies, and highly specific prompts that guide the diffusion model to produce more realistic conceptual renderings. We also conduct an experimental comparison between standard prompt-only generation and our RAG-based approach, measuring improvements in contextual accuracy and architectural suitability. The goal of this research is to show how AI can be used as an explainable, knowledge-backed assistant for early-stage conceptual design, rather than just a tool for generating aesthetic imagery.

Related Sustainable Development Goals (SDGs)

- **SDG 9: Industry, Innovation and Infrastructure** – Fosters innovation in architectural design workflows by integrating advanced AI and RAG technologies to support modern, resilient, and efficient infrastructure planning.
- **SDG 11: Sustainable Cities and Communities** – Promotes sustainable urban development by providing AI-driven recommendations for eco-friendly building designs, natural ventilation, and climate-specific architectural strategies.

- **SDG 12: Responsible Consumption and Production** – Encourages the efficient use of resources and low-energy consumption in building conceptualization through targeted material and sustainability recommendations.
- **SDG 13: Climate Action** – Integrates climate-aware reasoning into the architectural generation process, ensuring that building concepts are optimized for specific environmental conditions and contribute to climate adaptation.

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